

Amara

gives cordless drill 100

wants box of floor tiles 120

on Amara's scale

$$60/100 = 0.60$$

Bruno

gives box of floor tiles 50

wants cordless drill 60

on Bruno's scale

$$120/50 = 2.40$$

One barcode each: the things need no negotiation, only the rate does. Each rate is the taker's bid over the giver's ask, in mixed units. Around the loop the units cancel:

$$\prod \text{rate} = \prod_i \frac{\text{bid}_i}{\text{ask}_i} = \frac{120}{100} \cdot \frac{60}{50} = 1.44$$

Nothing is held or transferred on either scale.